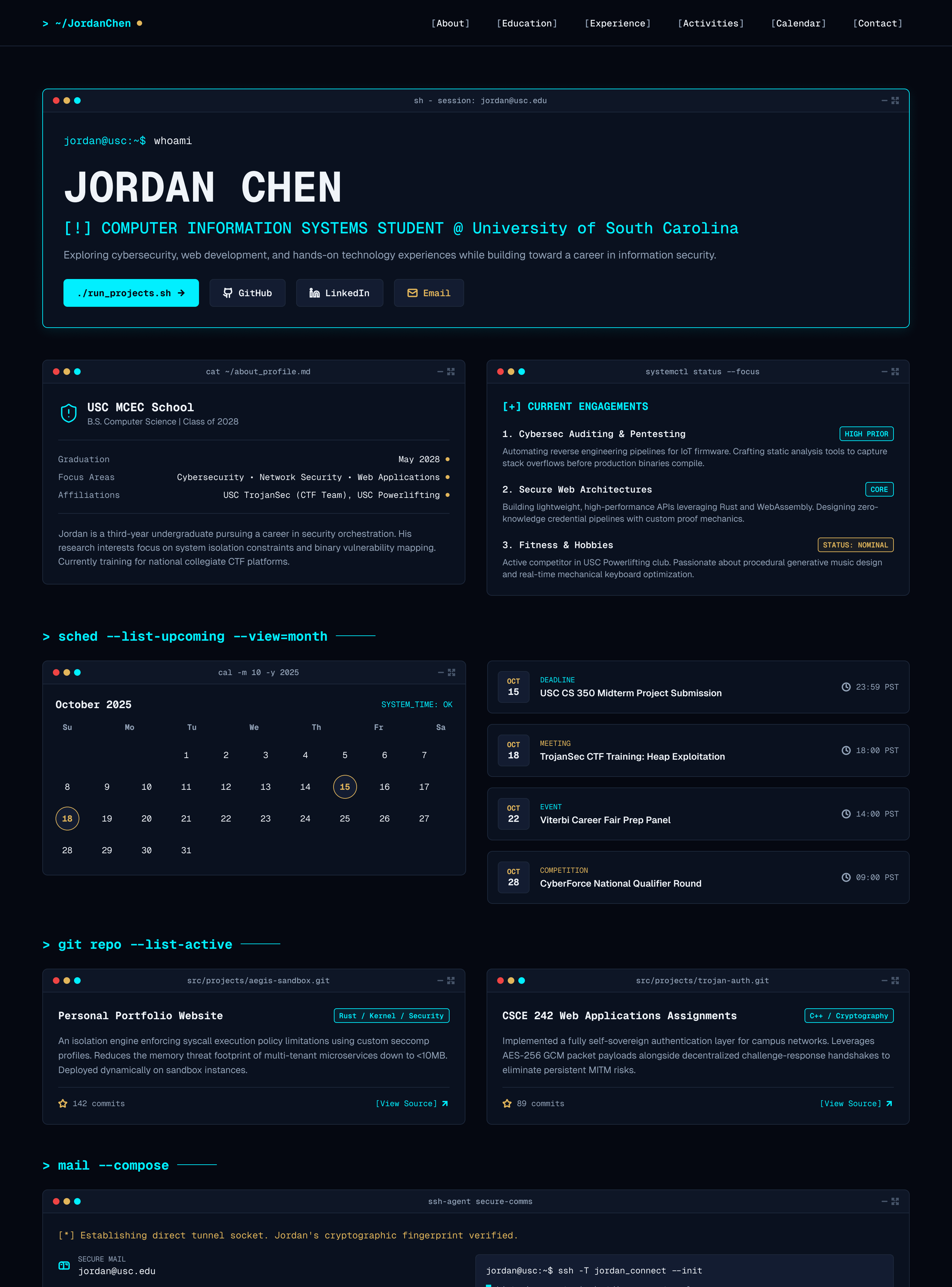

> ~/JordanChen ● [About] [Education] [Experience] [Activities] [Calendar] [Contact]

sh - session: jordan@usc.edu

`jordan@usc:~$ whoami`

JORDAN CHEN

[!] COMPUTER INFORMATION SYSTEMS STUDENT @ University of South Carolina

Exploring cybersecurity, web development, and hands-on technology experiences while building toward a career in information security.

cat ~/about_profile.md

USC MCEC School

B.S. Computer Science | Class of 2028

Graduation	May 2028 ●
Focus Areas	Cybersecurity • Network Security • Web Applications ●
Affiliations	USC TrojanSec (CTF Team), USC Powerlifting Club ●

Jordan is a third-year undergraduate pursuing a career in security orchestration. His research interests focus on system isolation controls and binary vulnerability mapping. Currently training for national collegiate CTF platforms.

systemctl status --focus

[+] CURRENT ENGAGEMENTS

1. Cybersec Auditing & Pentesting [HIGH PRIOR]
Automating reverse engineering pipelines for IoT firmware. Crafting static analysis tools to capture stack overflows before production binaries compile.

2. Secure Web Architectures [CORE]
Building lightweight, high-performance APIs leveraging Rust and WebAssembly. Designing zero-knowledge credential pipelines with custom proof mechanics.

3. Fitness & Hobbies [STATUS: NOMINAL]
Active competitor in USC Powerlifting club. Passionate about procedural generative music design and real-time mechanical keyboard optimization.

> sched --list-upcoming --view=month

cal -m 10 -y 2025

October 2025 SYSTEM_TIME: OK

```
Su  Mo  Tu  We  Th  Fr  Sa
            1   2   3   4
5   6   7   8   9   10  11
12  13  14  15  16  17  18
19  20  21  22  23  24  25
26  27  28  29  30  31
```

OCT 15 DEADLINE **USC CS 350 Midterm Project Submission** 23:59 PST

OCT 18 MEETING **TrojanSec CTF Training: Heap Exploitation** 18:00 PST

OCT 22 EVENT **Viterbi Career Fair Prep Panel** 14:00 PST

OCT 28 COMPETITION **CyberForce National Qualifier Round** 09:00 PST

> git repo --list-active

src/projects/aegis-sandbox.git

Personal Portfolio Website [Rust / Kernel / Security]

An isolation engine enforcing system execution policy limitations using custom seccomp profiles. Reduces the memory threat footprint of multi-tenant microservices down to <10MB. Deployed dynamically on sandbox instances.

☆ 142 commits [View Source] ↗

src/projects/trojan-auth.git

CSCE 242 Web Applications Assignments [C++ / Cryptography]

Implemented a fully self-sovereign authentication layer for campus networks. Leverages AES-256 GCM packet payloads alongside decentralized challenge-response handshakes to eliminate persistent MITM risks.

☆ 89 commits [View Source] ↗

> mail --compose

ssh-agent secure-comms

[*] Establishing direct tunnel socket. Jordan's cryptographic fingerprint verified.

SECURE MAIL
jordan@usc.edu

PGP PUBLIC KEY
0xFA3C8821BFF042D

```
jordan@usc:~$ ssh -T jordan_connect --init
█ Listening on standard stdin secure tunnels...
```

> Jordan Chen | B.S. CIS Student

● UPTIME: 99.98% SHELL: zsh STATUS: NOMINAL

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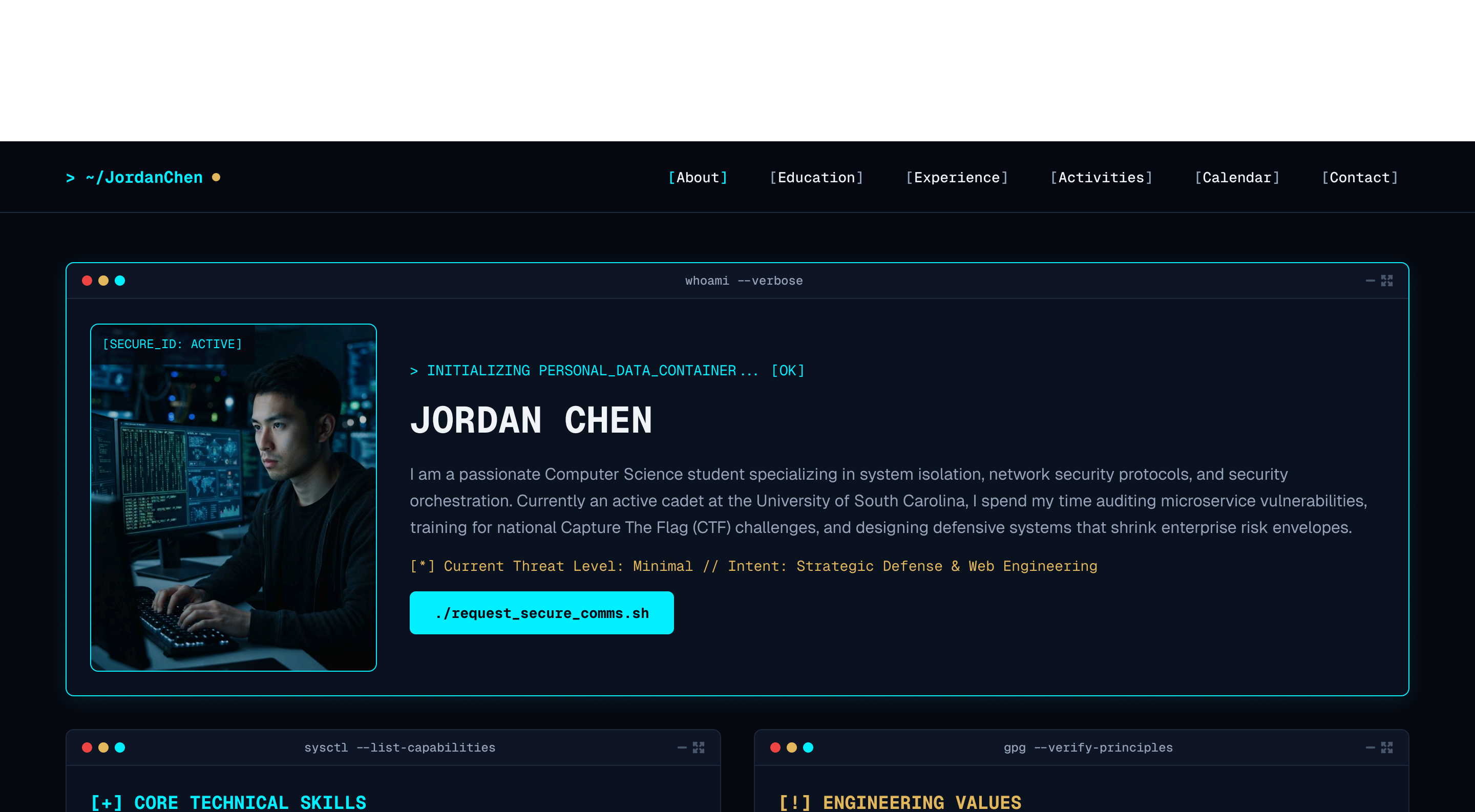
[github] [linkedin] [keybase]



> ~/JordanChen ● [About] [Education] [Experience] [Activities] [Calendar] [Contact]

whoami --verbose

[SECURE_ID: ACTIVE]

> INITIALIZING PERSONAL_DATA_CONTAINER... [OK]

JORDAN CHEN

I am a passionate Computer Science student specializing in system isolation, network security protocols, and security orchestration. Currently an active cadet at the University of South Carolina, I spend my time auditing microservice vulnerabilities, training for national Capture The Flag (CTF) challenges, and designing defensive systems that shrink enterprise risk envelopes.

[*] Current Threat Level: Minimal // Intent: Strategic Defense & Web Engineering

sysctl --list-capabilities

[+] CORE TECHNICAL SKILLS

Languages
Rust, C++, Python, JavaScript (ES6+), Assembly (x86_64), SQL

Security Tools
GDB, Ghidra, Wireshark, Burp Suite, Nmap, Metasploit, seccomp

Systems & Web
Linux (Debian, Arch), Docker, WebAssembly, Node.js, Express, AWS

gpg --verify-principles

[!] ENGINEERING VALUES

> Rigor
Security is built on strict isolation. Half-measures fail silently under modern adversarial patterns.

> Integrity
Operating with absolute clarity and legal adherence. Defending the integrity of public digital assets.

> Rethink
Systems must be engineered from zero-knowledge ground truths, ensuring defensive safety nets.

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● UPTIME: 99.98% SHELL: zsh STATUS: ACTIVE (ABOUT)

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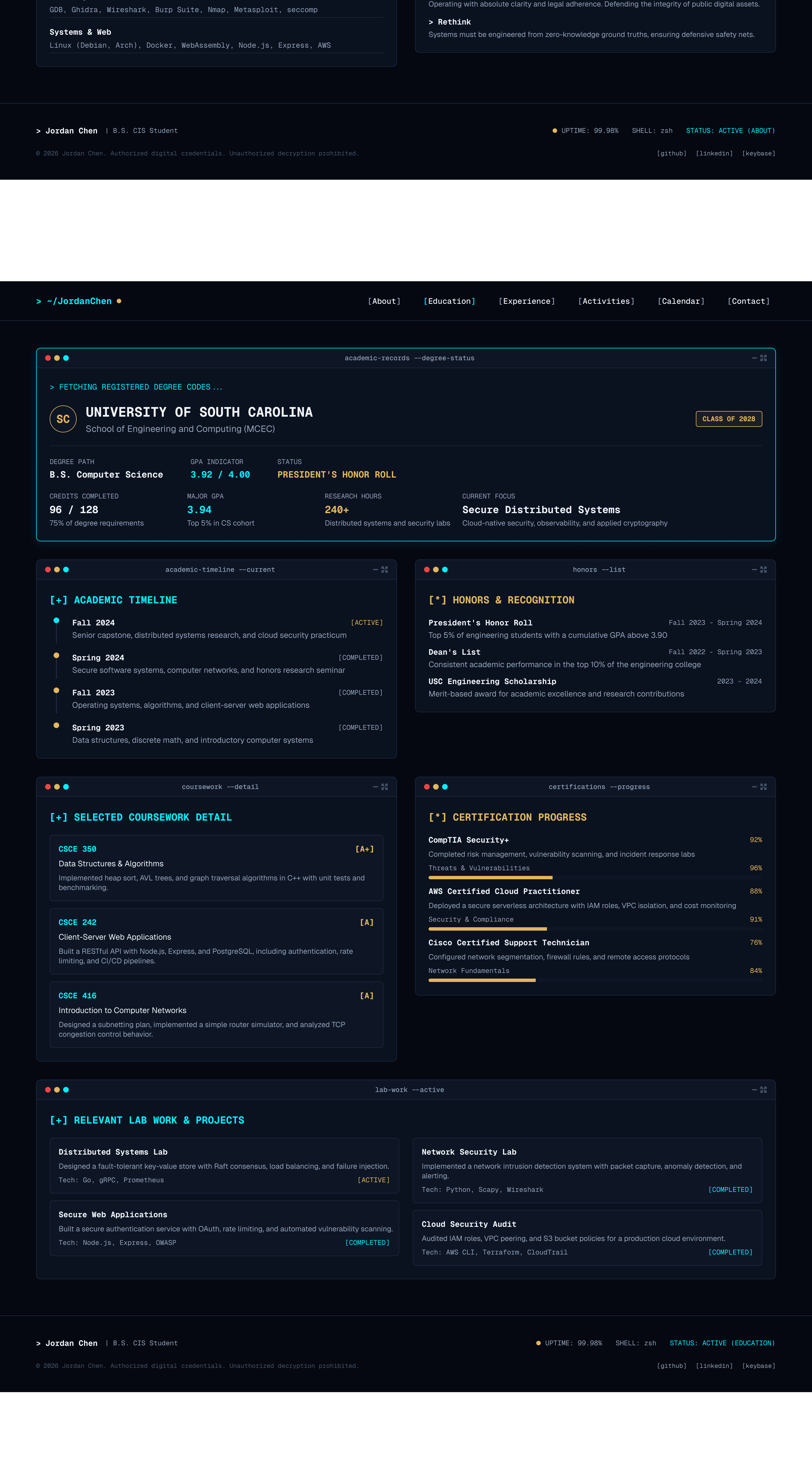
[github] [linkedin] [keybase]



> ~/JordanChen ● [About] [Education] [Experience] [Activities] [Calendar] [Contact]

academic-records --degree-status

> FETCHING REGISTERED DEGREE CODES...

SC **UNIVERSITY OF SOUTH CAROLINA**
School of Engineering and Computing (MCEC)

CLASS OF 2028

DEGREE PATH	GPA INDICATOR	STATUS	CURRENT FOCUS
B.S. Computer Science	3.92 / 4.00	PRESIDENT'S HONOR ROLL	Secure Distributed Systems Cloud-native security, observability, and applied cryptography
CREDITS COMPLETED	MAJOR GPA	RESEARCH HOURS	
96 / 128 75% of degree requirements	3.94 Top 5% in CS cohort	240+ Distributed systems and security labs	

academic-timeline --current

[+] ACADEMIC TIMELINE

- Fall 2024 [ACTIVE]
Senior capstone, distributed systems research, and cloud security practicum
- Spring 2024 [COMPLETED]
Secure software systems, computer networks, and honors research seminar
- Fall 2023 [COMPLETED]
Operating systems, algorithms, and client-server web applications
- Spring 2023 [COMPLETED]
Data structures, discrete math, and introductory computer systems

honors --list

[*] HONORS & RECOGNITION

President's Honor Roll — Fall 2023 - Spring 2024
Top 5% of engineering students with a cumulative GPA above 3.90

Dean's List — Fall 2022 - Spring 2023
Consistent academic performance in the top 10% of the engineering college

USC Engineering Scholarship — 2023 - 2024
Merit-based award for academic excellence and research contributions

coursework --detail

[+] SELECTED COURSEWORK DETAIL

CSCE 350 [A+]
Data Structures & Algorithms
Implemented heap sort, AVL trees, and graph traversal algorithms in C++ with unit tests and benchmarking.

CSCE 242 [A]
Client-Server Web Applications
Built a RESTful API with Node.js, Express, and PostgreSQL, including authentication, rate limiting, and CI/CD pipelines.

CSCE 416 [A]
Introduction to Computer Networks
Designed a subnetting plan, implemented a simple router simulator, and analyzed TCP congestion control behavior.

certifications --progress

[*] CERTIFICATION PROGRESS

CompTIA Security+ — 92%
Completed risk management, vulnerability scanning, and incident response labs
Threats & Vulnerabilities — 96%

AWS Certified Cloud Practitioner — 88%
Deployed a secure serverless architecture with IAM roles, VPC isolation, and cost monitoring
Security & Compliance — 91%

Cisco Certified Support Technician — 76%
Configured network segmentation, firewall rules, and remote access protocols
Network Fundamentals — 84%

lab-work --active

[+] RELEVANT LAB WORK & PROJECTS

Distributed Systems Lab
Designed a fault-tolerant key-value store with Raft consensus, load balancing, and failure injection.
Tech: Go, gRPC, Prometheus [ACTIVE]

Network Security Lab
Implemented a network intrusion detection system with packet capture, anomaly detection, and alerting.
Tech: Python, Scapy, Wireshark [COMPLETED]

Secure Web Applications
Built a secure authentication service with OAuth, rate limiting, and automated vulnerability scanning.
Tech: Node.js, Express, OWASP [COMPLETED]

Cloud Security Audit
Audited IAM roles, VPC peering, and S3 bucket policies for a production cloud environment.
Tech: AWS CLI, Terraform, CloudTrail [COMPLETED]

> Jordan Chen | B.S. CIS Student

● UPTIME: 99.98% SHELL: zsh STATUS: ACTIVE (EDUCATION)

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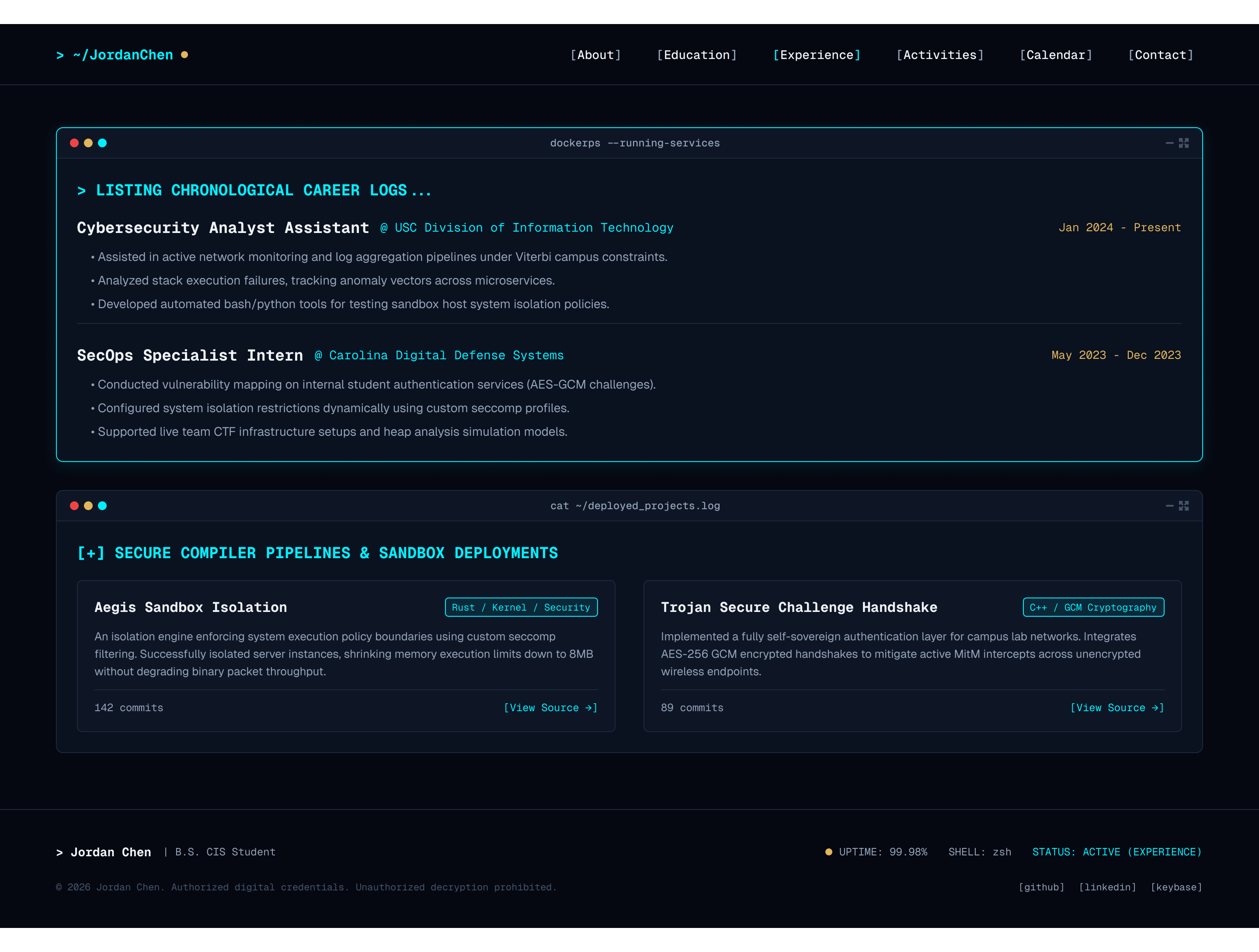
[github] [linkedin] [keybase]



> ~/JordanChen ● [About] [Education] [Experience] [Activities] [Calendar] [Contact]

dockerps --running-services

> LISTING CHRONOLOGICAL CAREER LOGS...

Cybersecurity Analyst Assistant @ USC Division of Information Technology — Jan 2024 - Present

- Assisted in active network monitoring and log aggregation pipelines under Viterbi campus constraints.
- Analyzed stack execution failures, tracking anomaly vectors across microservices.
- Developed automated bash/python tools for testing sandbox host system isolation policies.

SecOps Specialist Intern @ Carolina Digital Defense Systems — May 2023 - Dec 2023

- Conducted vulnerability mapping on internal student authentication services (AES-GCM challenges).
- Configured system isolation restrictions dynamically using custom seccomp profiles.
- Supported live team CTF infrastructure setups and heap analysis simulation models.

cat ~/deployed_projects.log

[+] SECURE COMPILER PIPELINES & SANDBOX DEPLOYMENTS

Aegis Sandbox Isolation [Rust / Kernel / Security]
An isolation engine enforcing system execution policy boundaries using custom seccomp filtering. Successfully isolated server instances, shrinking memory limits down to 8MB without degrading binary packet throughput.
142 commits [View Source ↗]

Trojan Secure Challenge Handshake [C++ / GCM Cryptography]
Implemented a fully self-sovereign authentication layer for campus lab networks. Integrates AES-256 GCM encrypted handshakes to mitigate active MITM intercepts across unencrypted wireless points.
89 commits [View Source ↗]

> Jordan Chen | B.S. CIS Student

● UPTIME: 99.98% SHELL: zsh STATUS: ACTIVE (EXPERIENCE)

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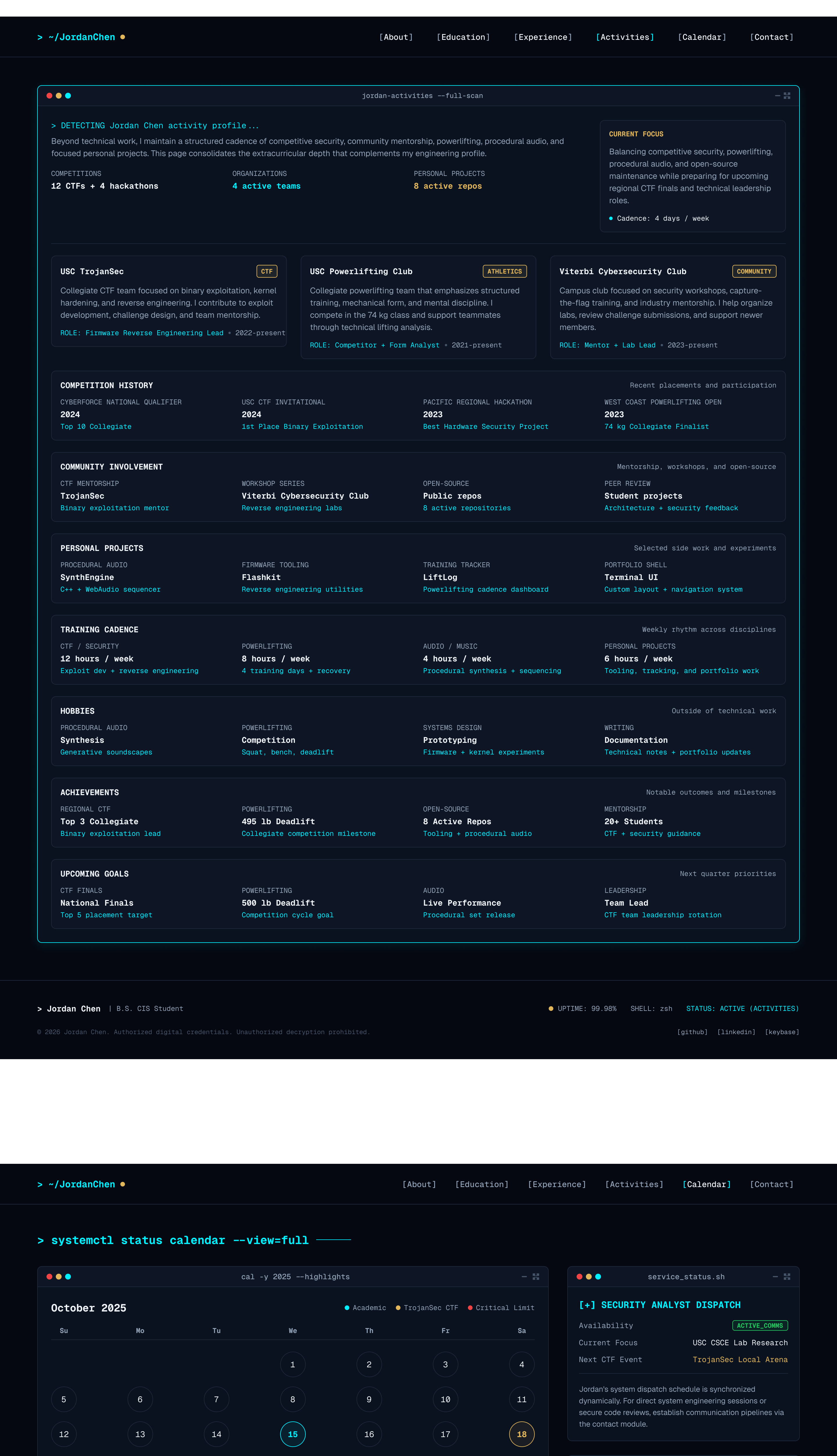
[github] [linkedin] [keybase]



> ~/JordanChen ● [About] [Education] [Experience] [Activities] [Calendar] [Contact]

jordan-activities --full-scan

> DETECTING Jordan Chen activity profile...

Beyond technical work, I maintain a structured cadence of competitive security, community mentorship, powerlifting, procedural audio, and open-source focused personal projects. This page consolidates the extracurricular depth that complements my engineering profile.

COMPETITIONS	ORGANIZATIONS	PERSONAL PROJECTS
12 CTFs + 4 hackathons	4 active teams	8 active repos

CURRENT FOCUS
Balancing competitive security, powerlifting, procedural audio, and open-source maintenance while preparing for upcoming regional CTF finals and technical leadership roles.
● Cadence: 4 days / week

USC TrojanSec [CTF]
Collegiate CTF team focused on binary exploitation, kernel hardening, and reverse engineering. I contribute to exploit development, challenge design, and team training.
ROLE: Firmware Reverse Engineering Lead • 2022-present

USC Powerlifting Club [ATHLETICS]
Collegiate powerlifting team that emphasizes structured training, recovery, and mental discipline. I compete in the 74 kg class and support teammates through technical lifting analysis.
ROLE: Competitor + Form Analyst • 2021-present

Viterbi Cybersecurity Club [COMMUNITY]
Campus club focused on security workshops, capture-the-flag training, and industry mentorship. I help organize labs, review challenge submissions, and support newer members.
ROLE: Mentor + Lab Lead • 2023-present

COMPETITION HISTORY — Recent placements and participation

CYBERFORCE NATIONAL QUALIFIER	USC CTF INVITATIONAL	PACIFIC REGIONAL HACKATHON	WEST COAST POWERLIFTING OPEN
2024	2024	2023	2023
Top 10 Collegiate	1st Place Binary Exploitation	Best Hardware Security Project	74 kg Collegiate Finalist

COMMUNITY INVOLVEMENT — Mentorship, workshops, and open-source

CTF MENTORSHIP	WORKSHOP SERIES	OPEN-SOURCE	PEER REVIEW
TrojanSec	Viterbi Cybersecurity Club	Public repos	Student projects
Binary exploitation mentor	Reverse engineering labs	8 active repositories	Architecture + security feedback

PERSONAL PROJECTS — Selected side work and experiments

PROCEDURAL AUDIO	FIRMWARE TOOLING	TRAINING TRACKER	PORTFOLIO SHELL
SynthEngine	Flashkit	LiftLog	Terminal UI
C++ + WebAudio sequencer	Reverse engineering utilities	Powerlifting cadence dashboard	Custom layout + navigation system

TRAINING CADENCE — Weekly rhythm across disciplines

CTF / SECURITY	POWERLIFTING	AUDIO / MUSIC	PERSONAL PROJECTS
12 hours / week	8 hours / week	4 hours / week	6 hours / week
Exploit dev + reverse engineering	4 training days + recovery	Procedural synthesis + sequencing	Tooling, tracking, and portfolio work

HOBBIES — Outside of technical work

PROCEDURAL AUDIO	POWERLIFTING	SYSTEMS DESIGN	WRITING
Synthesis	Competition	Prototyping	Documentation
Generative soundscapes	Squat, bench, deadlift	Firmware + kernel experiments	Technical notes + portfolio updates

ACHIEVEMENTS — Notable outcomes and milestones

REGIONAL CTF	POWERLIFTING	OPEN-SOURCE	MENTORSHIP
Top 3 Collegiate	495 lb Deadlift	8 Active Repos	20+ Students
Binary exploitation lead	Collegiate competition milestone	Tooling + procedural audio	CTF + security guidance

UPCOMING GOALS — Next quarter priorities

CTF FINALS	POWERLIFTING	AUDIO	LEADERSHIP
National Finals	500 lb Deadlift	Live Performance	Team Lead
Top 5 placement target	Competition cycle goal	Procedural set release	CTF team leadership rotation

> Jordan Chen | B.S. CIS Student

● UPTIME: 99.98% SHELL: zsh STATUS: ACTIVE (ACTIVITIES)

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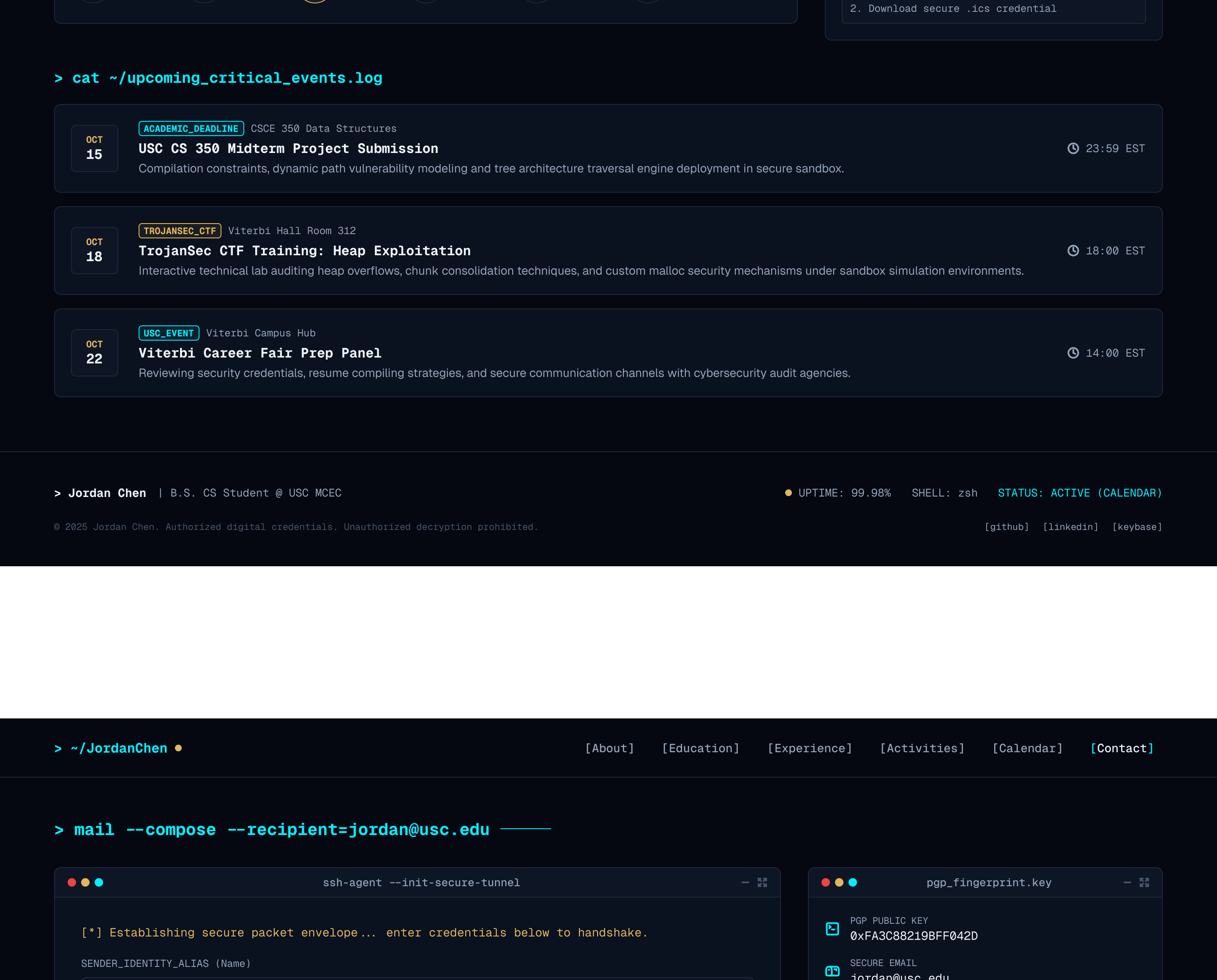
[github] [linkedin] [keybase]



> ~/JordanChen ● [About] [Education] [Experience] [Activities] [Calendar] [Contact]

> systemctl status calendar --view=full

cal -y 2025 --highlights

October 2025 ● Academic ● TrojanSec CTF ● Critical Limit

```
Su  Mo  Tu  We  Th  Fr  Sa
            1   2   3   4
5   6   7   8   9   10  11
12  13  14  15  16  17  18
19  20  21  22  23  24  25
26  27  28  29  30  31
```

service_status.sh

[+] SECURITY ANALYST DISPATCH

Availability	ACTIVE_COMMS
Current Focus	USC CSCE Lab Research
Next CTF Event	TrojanSec Local Arena

Jordan's system dispatch schedule is synchronized dynamically. For direct system engineering sessions or secure code reviews, establish communication pipelines via the contact module.

> query --sys-calendar

1. Sync local shell state (cron)
2. Download secure .ics credential

> cat ~/upcoming_critical_events.log

OCT 15 [ACADEMIC_DEADLINE] CSCE 350 Data Structures — 23:59 EST
USC CS 350 Midterm Project Submission
Compilation constraints, dynamic path vulnerability modeling and tree architecture traversal engine deployment in secure sandbox.

OCT 18 [TROJANSEC_CTF] Viterbi Hall Room 312 — 18:00 EST
TrojanSec CTF Training: Heap Exploitation
Interactive technical lab auditing heap overflows, chunk consolidation techniques, and custom malloc security mechanisms under sandbox simulation environments.

OCT 22 [USC_EVENT] Viterbi Campus Hub — 14:00 EST
Viterbi Career Fair Prep Panel
Reviewing security credentials, resume compiling strategies, and secure communication channels with cybersecurity audit agencies.

> Jordan Chen | B.S. CS Student @ USC MCEC

● UPTIME: 99.98% SHELL: zsh STATUS: ACTIVE (CALENDAR)

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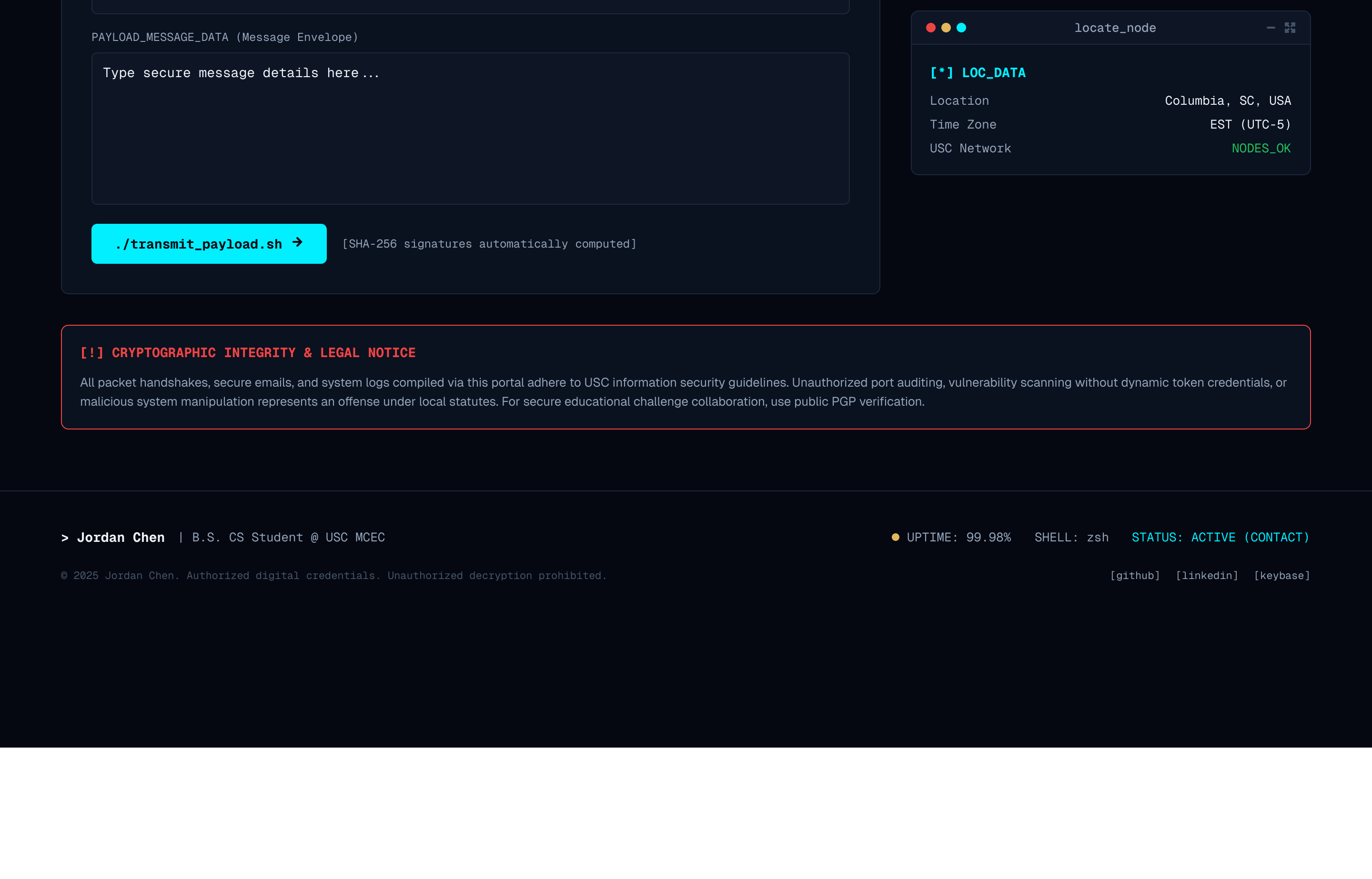
[github] [linkedin] [keybase]



> ~/JordanChen ● [About] [Education] [Experience] [Activities] [Calendar] [Contact]

> mail --compose --recipient=jordan@usc.edu

ssh-agent --init-secure-tunnel

[*] Establishing secure packet envelope... enter credentials below to handshake.

SENDER_IDENTITY_ALIAS (Name)
Enter Alias...

RETURN_ENDPOINT_ADDRESS (Secure Email)
Enter Return Email...

PAYLOAD_MESSAGE_DATA (Message Envelope)
Type secure message details here...

[SHA-256 signatures automatically computed]

pgp_fingerprint.key

PGP PUBLIC KEY
0xFA3C8821BFF042D

SECURE EMAIL
jordan@usc.edu

KEY_FINGERPRINT
FF30 A3C3 1E8B 82FC 44D9 FB21 8A54 E29B BBF0 C8E3

locate_node

[*] LOC_DATA

Location	Columbia, SC, USA
Time Zone	EST (UTC-5)
USC Network	NODES_OK

[!] CRYPTOGRAPHIC INTEGRITY & LEGAL NOTICE

All packet handshakes, secure emails, and system logs compiled via this portal adhere to USC information security guidelines. Unauthorized port auditing, vulnerability scanning without dynamic token credentials, or malicious system manipulation represents an offense under local statutes. For secure educational challenge collaboration, use public PGP verification.

> Jordan Chen | B.S. CS Student @ USC MCEC

● UPTIME: 99.98% SHELL: zsh STATUS: ACTIVE (CONTACT)

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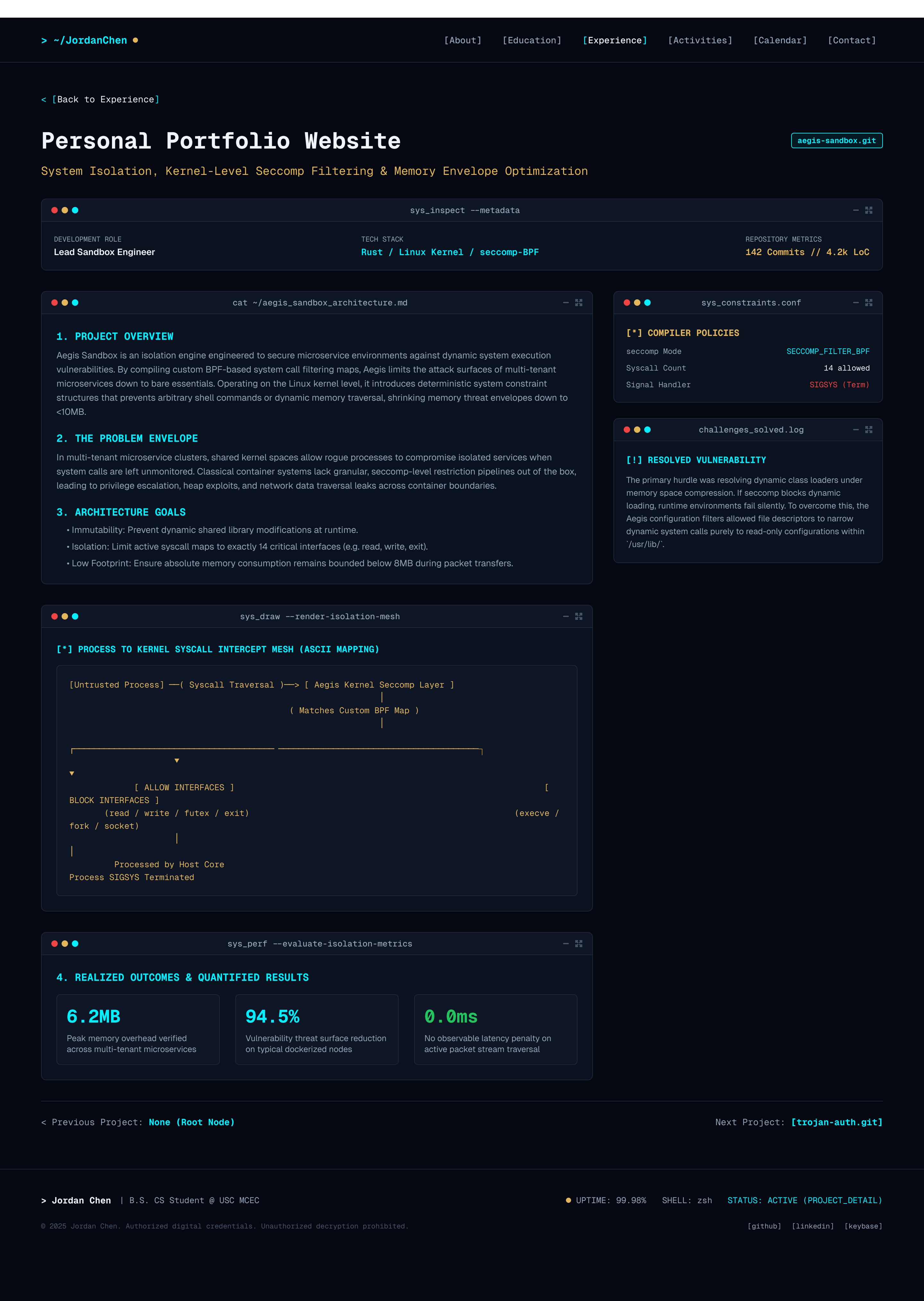
[github] [linkedin] [keybase]



> ~/JordanChen ● [About] [Education] [Experience] [Activities] [Calendar] [Contact]

< [Back to Experience]

Personal Portfolio Website

System Isolation, Kernel-Level Seccomp Filtering & Memory Envelope Optimization

aegis-sandbox.git

sys_inspect --metadata

DEVELOPMENT ROLE	TECH STACK	REPOSITORY METRICS
Lead Sandbox Engineer	Rust / Linux Kernel / seccomp-BPF	142 Commits // 4.2k LoC

cat ~/aegis_sandbox_architecture.md

1. PROJECT OVERVIEW

Aegis Sandbox is an isolation engine engineered to secure microservice environments against dynamic system execution vulnerabilities. By compiling custom BPF-based system call filtering maps, Aegis limits the attack surfaces of multi-tenant microservices down to bare essentials. Operating on the Linux kernel level, it introduces deterministic system constraint structures that prevents arbitrary shell commands or dynamic memory traversal, shrinking memory threat envelopes down to <10MB.

2. THE PROBLEM ENVELOPE

In multi-tenant microservice clusters, shared kernel spaces allow rogue processes to compromise isolated services when system calls are left unmonitored. Classical container systems lack granular, seccomp-level restriction pipelines out of the box, leading to privilege escalation, heap exploits, and network data traversal leaks across container boundaries.

3. ARCHITECTURE GOALS

- Immutability: Prevent dynamic shared library modifications at runtime.
- Isolation: Limit active syscall maps to exactly 14 critical interfaces (e.g. read, write, exit).
- Low Footprint: Ensure absolute memory consumption remains bounded below 8MB during packet transfers.

sys_constraints.conf

[*] COMPILER POLICIES

seccomp Mode	SECCOMP_FILTER_BPF
Syscall Count	14 allowed
Signal Handler	SIGSYS (Term)

challenges_solved.log

[!] RESOLVED VULNERABILITY

The primary hurdle was resolving dynamic class loaders under memory space compression. If seccomp blocks dynamic loading, runtime environments fail silently. To overcome this, the Aegis configuration filters allowed file descriptors to narrow dynamic system calls purely to read-only configurations within `/usr/lib/`.

sys_draw --render-isolation-mesh

[*] PROCESS TO KERNEL SYSCALL INTERCEPT MESH (ASCII MAPPING)

```
[Untrusted Process] ──( Syscall Traversal )──> [ Aegis Kernel Seccomp Layer ]
                                                        |
                                            ( Matches Custom BPF Map )
                                                        |

┌──────────────────────────────────────── ────────────────────────────────────────┐
                    ▼
▼
             [ ALLOW INTERFACES ]                                                  [
BLOCK INTERFACES ]
       (read / write / futex / exit)                                           (execve /
fork / socket)
                    |
|
         Processed by Host Core
Process SIGSYS Terminated
```

sys_perf --evaluate-isolation-metrics

4. REALIZED OUTCOMES & QUANTIFIED RESULTS

6.2MB	94.5%	0.0ms
Peak memory overhead verified across multi-tenant microservices	Vulnerability threat surface reduction on typical dockerized nodes	No observable latency penalty on active packet stream traversal

< Previous Project: **None (Root Node)**

Next Project: **[trojan-auth.git]**

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● UPTIME: 99.98% SHELL: zsh STATUS: ACTIVE (PROJECT_DETAIL)

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[github] [linkedin] [keybase]